Ammar Zaghloul

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ammarz.com | github.com/ammar-az | linkedin.com/in/ammar-zaghloul

Skills

Languages	C#, TypeScript, JavaScript, SQL, Python, C++, Java, HTML/CSS
Frameworks	ASP.NET Core, Entity Framework Core, React, React Router, Hono, Tailwind CSS
APIs & Runtime	REST APIs, JWT authentication, refresh tokens, WebSockets, Node.js
Databases	SQL Server, Azure SQL Database, Cloudflare D1 (SQLite), MySQL
Tools & Cloud	Git, GitHub, Docker, Azure App Service, Cloudflare Workers, Durable Objects, Vite

Projects

Tasked — Project Management Application

C# | ASP.NET Core | EF Core | SQL Server | React | TypeScript | Docker | Azure

tasked.ammarz.com | Frontend repository | Backend repository

- Built a collaborative project and task management application with memberships, invitations, and role-based authorization.
- Designed relational data models and query endpoints for task search, status and assignee filters, sorting, and pagination.
- Implemented JWT sessions with memory-held access tokens, HttpOnly refresh-token cookies, hashed refresh-token storage, and token rotation.
- Deployed the containerized API and database to Azure App Service and Azure SQL, hosted the frontend on Cloudflare Workers, and managed schema changes through migrations.

LoL Theme Guess — Real-Time Multiplayer Web Game

React | TypeScript | Tailwind CSS | Hono | Cloudflare Workers | Durable Objects | D1 (SQLite)

lol-theme-guess.ammarz.com

- Built a solo and multiplayer game where players identify League of Legends champions from audio snippets.
- Isolated multiplayer lobbies in Durable Objects, using WebSocket broadcasts and server-side alarms to coordinate rounds, validate guesses, and update scores.
- Implemented skip voting, join/disconnect updates, and lobby cleanup; retrieved theme metadata with parameterized D1 queries.

Education

University of Calgary — *Calgary, Canada*

Bachelor of Science with Distinction in Computer Science | GPA: 3.7/4.0 — Sept 2020 – Feb 2025

Research Experience

Undergraduate Research Assistant — *University of Calgary*

Serious Games Research Group — May 2024 – Aug 2024

- Secured NSERC USRA funding through a VR learning proposal and independently developed the proof-of-concept application in Unreal Engine.
- Conducted a usability study to assess the VR environment's effectiveness and user engagement.